CSE221: Algorithms
Lab-03
Summer 2026

Action required for Students

[Google Form] CSE221 Lab Policy:
- The lab policy file can be found here - Link
- Please complete the lab policy acknowledgement form - Link

[Google Form] Codeforces Account Information: (Action required)
- Open an account at https://codeforces.com/
- Please submit your codeforces details here - Link

Students who have not completed this yet, please do so ASAP. Otherwise, your assessments will not be graded and Negative marks will be displayed on the lab marks portal.

Quiz 03

Date: July 11-12, 2026 (as per your lab class schedule)
Syllabus: Lab 03 content

Take Preparation for both assignment problems and outside assignment problems
[Student Drive Folder Link]

Assignment-03 Links:

codeforces Invitation Link:
https://codeforces.com/contestInvitation/b56aedf4f65aa3997c518df6682d6d1670a70079

Alternative Link: **https://codeforces.com/contests/701904**
[The alternative link will **only work** after you have successfully registered using the first link.]

Deadline: `Jul 12, 2026 2:00 PM`

Note: Use FAST I/O in your code to avoid Time Limit Exceeded (TLE), especially in the quiz.

After the deadline ends, you can submit your solution by entering the contest as a Virtual Participant for practice purposes. **However, please note, submissions after the deadline are not accepted for grading.**

Please note that your assignment submissions will not be evaluated if they are flagged as AI-generated.

Learning Outcome:

In this lab, we will explore the **Divide and Conquer** paradigm, which is a fundamental algorithmic technique for solving problems efficiently by breaking them into smaller subproblems. Through hands-on problem-solving, students will develop a deeper understanding of recursion, algorithm design, and optimization.

Understanding the Divide and Conquer Approach

- Learn how to break a problem into **smaller independent subproblems**, solve them recursively, and combine the results.
- Understand when and why **Divide and Conquer** is preferable over brute-force approaches.
- Apply recurrence relations and the Master Theorem to analyze **time complexity.**